Table 8: Deployment data composition. **Type**: agentic (A), chat (C), or mixed (M). For mixed categories the agentic/chat split is given in parentheses. OpenHands and Open Deep Research are retained from their respective agentic corpora at their native sample counts.

Source	Origin / notes	Type	n
dataclaw [O'Malley, 2026]	Agentic CLI sessions; 29 generator models across 8 families, 5 scaffolds	M (103/12)	115
reddit_wayback	Scraped public chat	C	56
wildbench [Lin et al., 2024]	Benchmark prompts (reasoning/math/data_analysis, 10 each)	C	30
hh_rlhf_redteam [Bai et al., 2022]	Anthropic red-team prompts	C	20
toxic_chat [Lin et al., 2023, Gehman et al., 2020]	Toxic-Chat + Real Toxicity Prompts (downsampled for balance)	C	20
author_chats	ChatGPT / Gemini share links	C	18
codechat [Zhong et al., 2025]	CodeChat dataset	C	18
cdial_bias [Zhou et al., 2022]	CDial-Bias dialogue	C	15
traced_run [Schaeffer, 2026]	Dataclaw agentic traces with operator-flagged misbehaviour (fixed-width window)	M (12/1)	13
open_deep_research [Roucher et al., 2025]	Retained agentic corpus	A	10
crowd_sourced_code [Yan et al., 2026]	Community-donated CLI sessions (incl. merged claude_code records)	A	10
openhands [Wang et al., 2024]	Retained agentic corpus (SWE-Gym + OpenHands)	A	8

H Model list

Table 9: Overview of evaluated models.

Provider	API model ID	Released
Anthropic	`claude-haiku-4-5`	Oct 2025
Anthropic	`claude-opus-4-6`	Feb 2026
Anthropic	`claude-sonnet-4-6`	Feb 2026
OpenAI	`gpt-4.1`	Apr 2025
OpenAI	`gpt-5.1`	Nov 2025
OpenAI	`gpt-5.4`	Mar 2026
Google DeepMind	`gemini-2.5-pro`	Jun 2025
Google DeepMind	`gemini-3.1-pro-preview`	Feb 2026
DeepSeek	`deepseek-v3.2`	Dec 2025
Z.ai	`glm-5.1`	Mar 2026
Moonshot AI	`kimi-k2.5`	Jan 2026
Alibaba	`qwen3.5-397b-a17b`	Feb 2026

I Methodological Corrections on AUROC

Figure 12 compares four pipeline configurations: the fixed Needham et al. probe set, per-model probe selection under mixed deployment data, corrected mixed-deployment AUROC, and a self-only reference using same-family deployment data.

Per-model PQ selection (Mixed vs. Baseline) lifts AUROC by 0.00–0.08 across judges, and the stratified correction (Corrected vs. Mixed) adds a further 0.03–0.07. Both lifts are uneven: three judges gain nothing from PQ selection while three gain meaningfully, and the correction's effect varies similarly across judges. The Corrected column tracks the self-only reference closely AUROC for all six judges), supporting the validation in Section 3.3.3 that harmonisation recovers most of the unconfounded signal without requiring same-family deployment data.

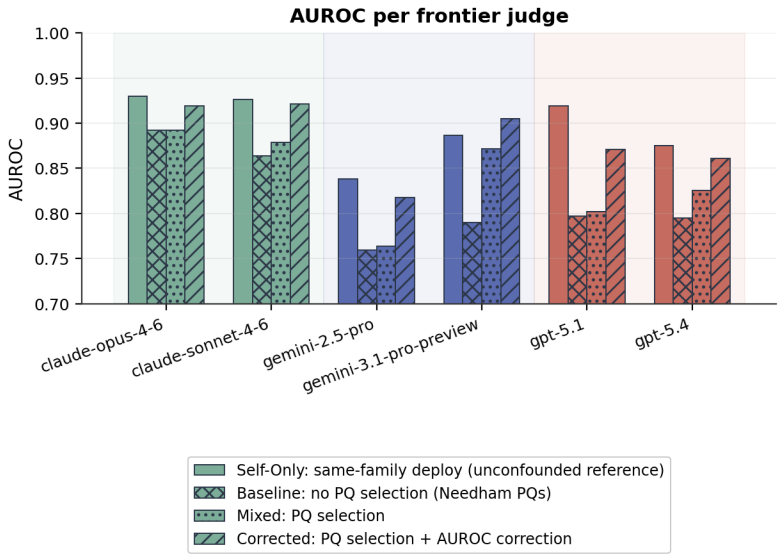


Figure 12: **AUROC per frontier judge across four pipeline configurations. Baseline** uses Needham et al.'s fixed probe-question set on the heterogeneous deploy pool. **Mixed** applies per-model probe selection (Section 3.2). **Corrected** adds the stratified harmonisation correction (Section 3.3.3) on top. **self-only** restricts deployment data to the judge's own model family, eliminating the generator-identity confound at the cost of sample size and diversity. All conditions evaluated on the same transcripts (self-only uses a subset). Judges ordered by family; shading denotes family membership.

J Compute and API Cost for Replicating This Work

Table 10 reports estimated API cost per experiment. Costs are derived from token counts $\times$ provider list prices (Jan 2026): per judge call we charge for the system prompt, the full transcript, the probe-question template, and a $\sim$200-token response, multiplied by the number of probes used in that experiment (1 for the 6$\times$6 deploy heatmap, $k \in \{1, 3, 5, 9\}$ for the depth resample, 5 for the cross-benchmark measure suite, 1 for Appendix B). Generator costs are cumulative across multi-turn rollouts.

A single tokenizer (`cl100k_base`) is used for all models; the $\pm 10\%$ band absorbs per-provider tokenizer disagreement (Anthropic $\sim +7\%$, Google $\sim -10\%$, others within $\pm 5\%$ on our text). Batch columns apply the standard 50% provider discount. In our runs only the Anthropic judges were batched, putting the realised replication cost between the two columns at $2,627. The PQ-selection sweep (Figure 7) is listed separately as it was not part of this replication.

Table 10: Estimated API spend per experiment. Non-batch and batch columns assume all calls at list price and all calls at the standard 50% batch discount. Realised replication cost falls between the two, at $2,627 with Anthropic-only batching. Estimates use Jan 2026 list prices and do not model prompt caching; caching reduces Anthropic-side input by 30–40% on multi-PQ runs.

Experiment	Non-batch (USD)		Batch – 50% off (USD)	
	avg	range (±10%)	avg	range (±10%)
Generator collection (transcript suite)	$237	$214–$261	$119	$107–$130
6×6 deploy heatmap	$213	$192–$234	$106	$96–$117
Depth resample	$1,968	$1,771–$2,164	$984	$885–$1,082
Cross-benchmark measure	$1,763	$1,587–$1,939	$882	$793–$970
Style isolation, Cond. A (App. B)	$108	$97–$119	$53.90	$48.51–$59.30
Style isolation, Cond. B (App. B)	$43.03	$38.73–$47.34	$21.52	$19.37–$23.67
Retry surcharge (measure)	$36.82	$33.14–$40.50	$18.41	$16.57–$20.25
Replication subtotal	**$4,369**	$3,932–$4,805	**$2,184**	$1,966–$2,403
PQ-selection sweep	$4,642	$4,178–$5,106	$2,321	$2,089–$2,553
Grand total	**$9,011**	$8,110–$9,912	**$4,505**	$4,055–$4,956

K Cost Estimate for a New Benchmark

A practitioner applying EvalDetectBench to their own benchmark faces two separable costs:

$$C \;\approx\; C_{\text{gen}} + C_{\text{probe}} \tag{3}$$

The first term, C_{gen}, is the cost of generating the transcripts to be scored. This is the subject model producing trajectories on the target benchmark. This term is zero for most adopters: if the target benchmark has already been run, the transcripts already exist and have already been paid for. We therefore treat C_{gen} as a term equal to the cost of running the target benchmark once, which the practitioner can estimate from their own benchmark.

The second term, C_{probe}, is the probe pass: scoring each transcript with the calibrated probe questions to read out $P(\text{eval})$. The probe pass has two stages: calibration and scoring. During the calibration stage, the full corpus of K probe questions is run to select the top-k performers on a calibration set comprising a fraction f of the transcripts. During the scoring stage, only the selected k probes are run on the remaining $(1 - f)$ of the transcripts. The probe cost is therefore the sum of a calibration term and a scoring term:

$$C_{\text{probe}} \;\approx\; C_{\text{calib}} + C_{\text{score}}, \tag{4}$$

$$C_{\text{calib}} \;\approx\; f\, N_t \cdot K \cdot \big(c_{\text{in}}(o + \bar{T}) + c_{\text{out}}\, r\big), \tag{5}$$

$$C_{\text{score}} \;\approx\; (1 - f)\, N_t \cdot k \cdot \big(c_{\text{in}}(o + \bar{T}) + c_{\text{out}}\, r\big), \tag{6}$$

where N_t is the total number of transcripts, f is the calibration fraction, K is the full probe-question corpus size, k is the number of selected probe questions, c_{in} and c_{out} are the probe model's per-token input and output prices, o is the fixed system-prompt and wrapper overhead per probe, $\bar{T}$ is the mean transcript length in tokens, and r is the mean response length per judgment. The per-call cost is identical across both stages; only the number of probe passes differs.